

Flow transition over surface gaps in 2D incompressible laminar boundary layers

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Motivation



Figure: Wing of a Boeing 737-800

Setup

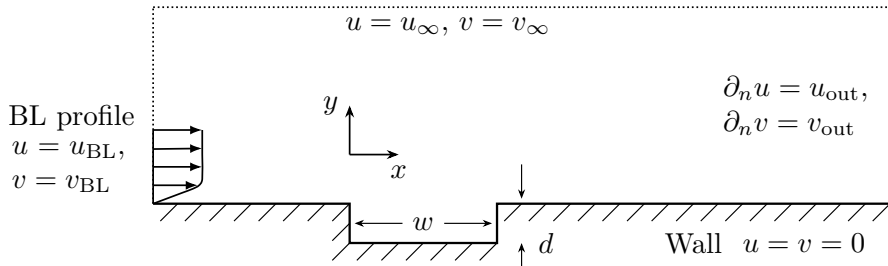


Figure: Domain setup for the steady-state finder

- **Aim:** Study the stability of the system as a function of the depth d and width w of the gap.
- 2D incompressible NS
- $Re_{\delta^*} = 1000 \implies Re_x = 3.38 \times 10^5$

Stability results

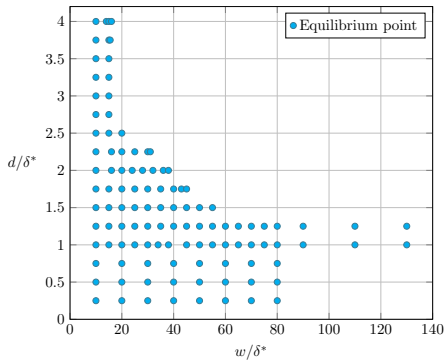


Figure: Classification of the stability of points downstream of the gap.

Stability results

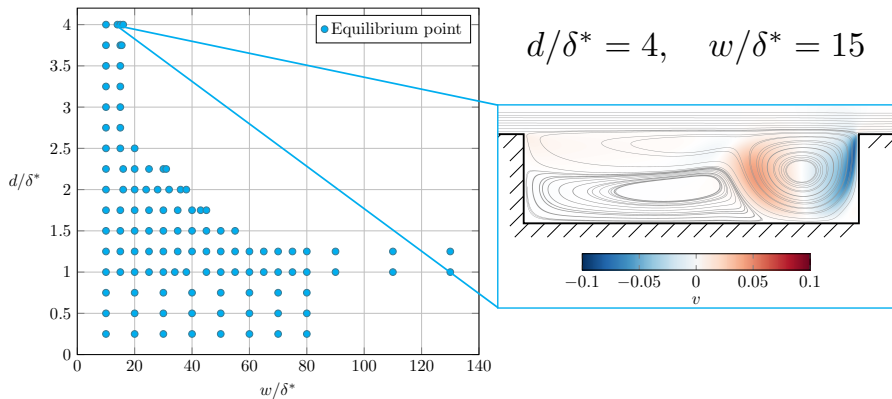


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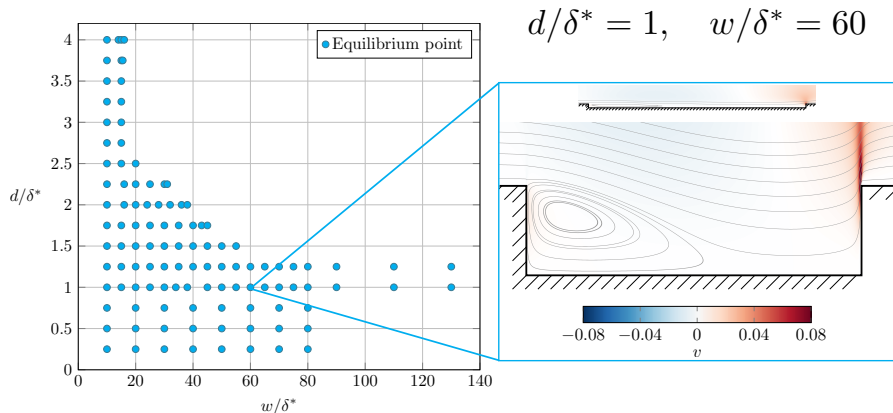


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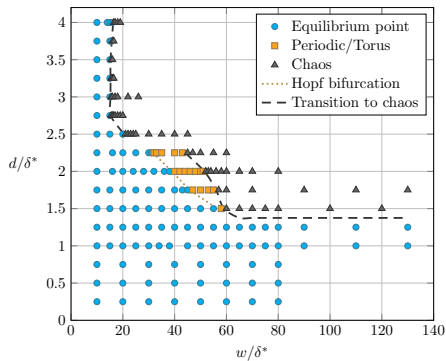


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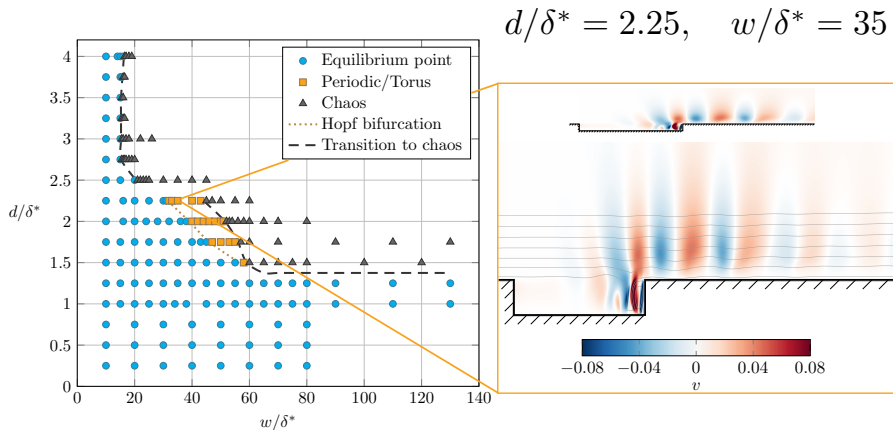


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Framework for the LST (TS-wave transition)

- We linearize the flow around a steady baseflow:

$$\mathbf{u}(x, y, t) = \mathbf{U}(x, y) + \tilde{\mathbf{u}}(x, y, t).$$

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- We add a disturbance of characteristic amplitude A_0 upstream of the gap and we track its evolution through

$$A(x) := A_0 e^{n(x)} \implies n(x) = \log \left(\frac{A(x)}{A_0} \right).$$

$n(x)$ is called the *n-factor* and it is used to predict transition in the BL.

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- To compare between gaps, we define $n_{\text{ref}}(x)$ as the *n-factor* of a reference case (flat plate) and we compute

$$\Delta n_{\text{gap}} = n_{\text{gap}} - n_{\text{ref}}.$$

Perturbed system setup

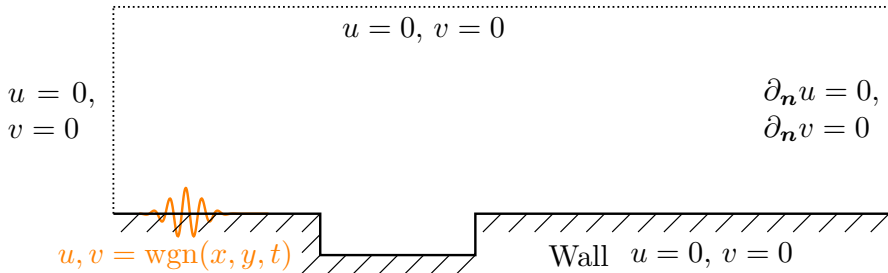


Figure: Domain setup for the perturbed system

Previous Work

Flow (2022), 2 E8
doi:10.1017/fo.2022.1

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Characterizing surface-gap effects on boundary-layer transition dominated by Tollmien–Schlichting instability

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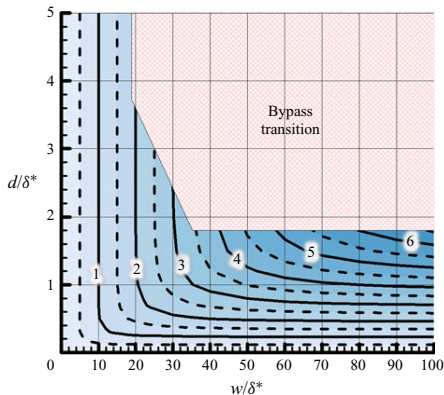


Figure: $\Delta n_{\text{gap}} = n_{\text{gap}} - n_{\text{ref}}$ for different gap dimensions

Crouch JD, Kosorygin VS, Sutanto MI, Miller GD. Characterizing surface-gap effects on boundary-layer transition dominated by Tollmien–Schlichting instability. *Flow*. 2022;2:E8.

e^N -method results

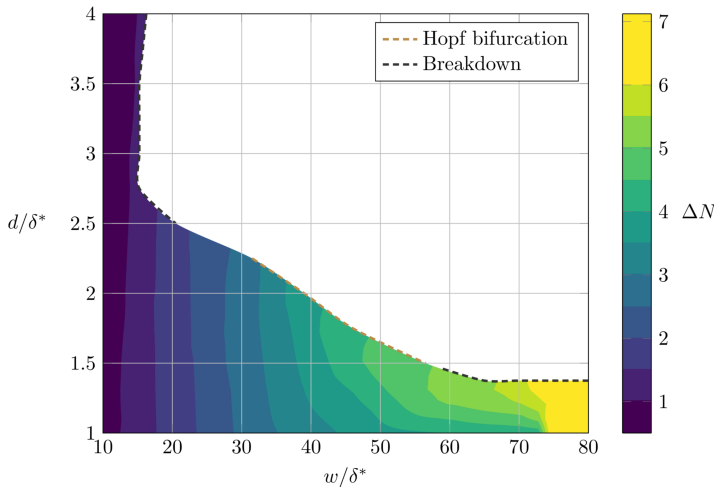


Figure: Interpolated $\Delta n_{\text{gap}} = n_{\text{gap}} - n_{\text{ref}}$ in the globally-stable region.

e^N -method results

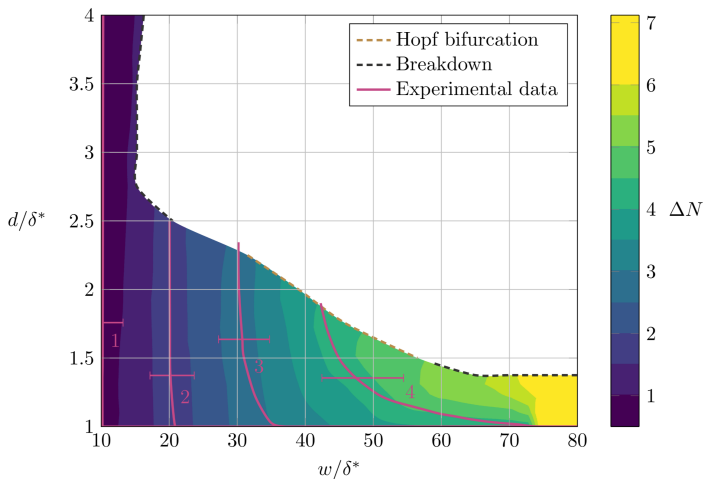


Figure: Interpolated $\Delta n_{\text{gap}} = n_{\text{gap}} - n_{\text{ref}}$ in the equilibria region. Magenta lines indicate the contour levels of the experimental data.

Future Work

- Go to higher Ma (compressible regime).

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- Account for spanwise effects (quasi-3d simulations).